

Experiment – 9

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Design Input Forms with Validation and Error Messages

INTRODUCTION

This experiment focuses on designing a **user input form** that validates data such as **email** and **phone number** and displays appropriate **error messages**. The goal is to ensure that users enter correct and valid information before submission.

FORM DESIGN OVERVIEW

The form is created using:

- **HTML** → Structure of the form

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>Form Validation</title>
7      <link rel="stylesheet" href="style.css">
8  </head>
9  <body>
10     <div class="container">
11         <form id="myForm">
12             <label for="email">Email:</label>
13             <input type="email" id="email" name="email" required>
14
15             <span id="emailError" class="error"></span>
16
17             <label for="phone">Phone Number:</label>
18             <input type="text" id="phone" name="phone" required>
19             <span id="phoneError" class="error"></span>
20
21             <button type="submit">Submit</button>
22         </form>
23     </div>
24     <script
25         src="https://cdnjs.cloudflare.com/ajax/libs/validate.js/0.13.1/validate.js"
26         <script src="script.js"></script>
27 </body>
28 </html>
```

- **CSS** → Styling and layout

```

# style.css > ...
1  body {
2    font-family: Arial,sans-serif;
3    background-color: #f4f4f4;
4    display: flex;
5    justify-content: center;
6    align-items: center;
7    height: 100vh;
8    margin: 0;
9  }
10
11  .container {
12    background-color: white;
13    padding: 20px;
14    border-radius: 5px;
15    box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
16  }
17
18  form {
19    display: flex;
20    flex-direction: column;
21  }
22
23  label {
24    margin-bottom: 5px;
25  }
26
27  input {
28    margin-bottom: 10px;
29    padding: 10px;
30    border: 1px solid #ccc;
31    border-radius: 3px;
32  }
33
34  button {
35    padding: 10px;
36    background-color: #28a745;
37    color: white;
38    border: none;
39
40    border-radius: 3px;
41    cursor: pointer;
42  }
43
44  button:hover {

```

- JavaScript → Validation logic

```

# script.js > ...
1  document.getElementById('myForm').addEventListener('submit',
2    e.preventDefault();
3
4    let email = document.getElementById('email').value;
5    let phone = document.getElementById('phone').value;
6    let emailError = document.getElementById('emailError');
7    let phoneError = document.getElementById('phoneError');
8    // Clear previous error messages
9    emailError.textContent = '';
10
11    phoneError.textContent = '';
12    // Validate email
13    if (!validator.isEmail(email)) {
14      emailError.textContent = 'Please enter a valid email';
15    }
16    // Validate phone number
17    if (!validator.isMobilePhone(phone, 'any')) {
18      phoneError.textContent = 'Please enter a valid phone number';
19    }
20
21    // If no errors, submit the form (for demonstration purposes only)
22    if (validator.isEmail(email) && validator.isMobilePhone(phone)) {
23      console.log('Email:', email);
24      console.log('Phone:', phone);
25    }
26  });

```

INPUT FIELDS

1. Email Field

- Accepts user email input
- Validation checks:
 - Must contain “@” symbol
 - Must follow proper email format

2. Phone Number Field

- Accepts numeric input
- Validation checks:
 - Must be a valid phone number
 - Ensures correct format (digits only)

VALIDATION PROCESS

- When the user clicks **Submit**, JavaScript:
 1. Prevents default form submission
 2. Retrieves input values
 3. Validates email and phone number using conditions
 4. Displays error messages if inputs are invalid

ERROR HANDLING

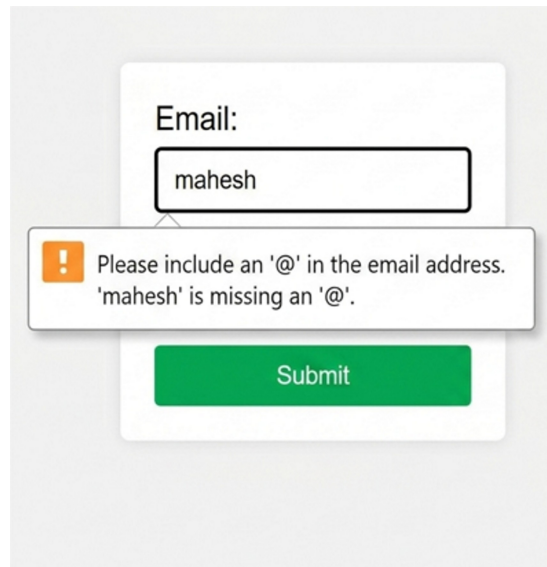
- Error messages are shown below each input field
- Example (from your output):
 - Invalid Email:
“Please include an ‘@’ in the email address”
- If inputs are correct:
 - Data is accepted and logged successfully

UI DESIGN FEATURES

- Centered form layout
- Clean and minimal design
- Highlighted **Submit button**
- Error messages displayed in a clear manner

OUTPUT ANALYSIS

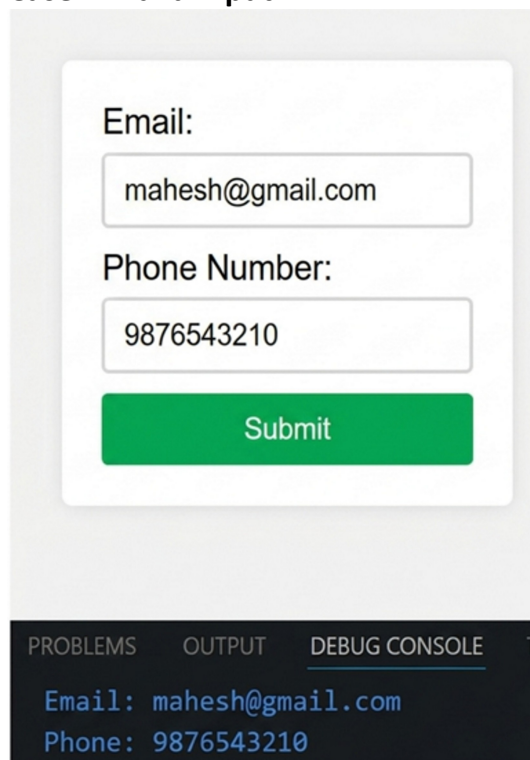
Case 1: Invalid Input



A web form with a label "Email:" and a text input field containing "mahesh". Below the input field is an orange error message box with a white exclamation mark icon. The message reads: "Please include an '@' in the email address. 'mahesh' is missing an '@'." Below the error message is a green "Submit" button.

- Email without “@” shows error message
- Prevents form submission

Case 2: Valid Input



A web form with two labels: "Email:" and "Phone Number:". The "Email:" input field contains "mahesh@gmail.com" and the "Phone Number:" input field contains "9876543210". Below the inputs is a green "Submit" button. At the bottom of the form is a dark grey debug console with three tabs: "PROBLEMS", "OUTPUT", and "DEBUG CONSOLE". The "DEBUG CONSOLE" tab is active and shows the following text: "Email: mahesh@gmail.com" and "Phone: 9876543210".

- Correct email and phone number accepted
- No error messages displayed

CONCLUSION

This experiment demonstrates how **form validation improves usability and data integrity**. By using JavaScript validation and clear error messages, the form ensures that users provide correct information before submission.